

WASP-2b

R-0253

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-2_240902 · created 2026-07-17T06:52:52+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460555.7864 +/- 0.0061 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.129 +/- 0.034
Transit depth (Rp/Rs)^2	1.67 +/- 0.87 [%]
Orbital Inclination (inc)	89.12 +/- 2.13
Airmass coefficient 1 (a1)	0.981 +/- 0.011
Airmass coefficient 2 (a2)	0.011 +/- 0.019
Scatter in the residuals of the lightcurve fit is	1.09 %
Best Comparison Star	#3 - [495, 63]
Optimal Aperture	3.68
Optimal Annulus	7.56
Transit Duration (day)	0.0626 +/- 0.0073

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.129 ± 0.034 vs 0.1326 (deviation 0.11σ)
Duration fit vs published	0.0626 d vs 0.0737 d (deviation 1.53σ)
Mid-time O–C	-8.6 min
Depth SNR	3.8
Residual scatter	1.09 %

Light curve

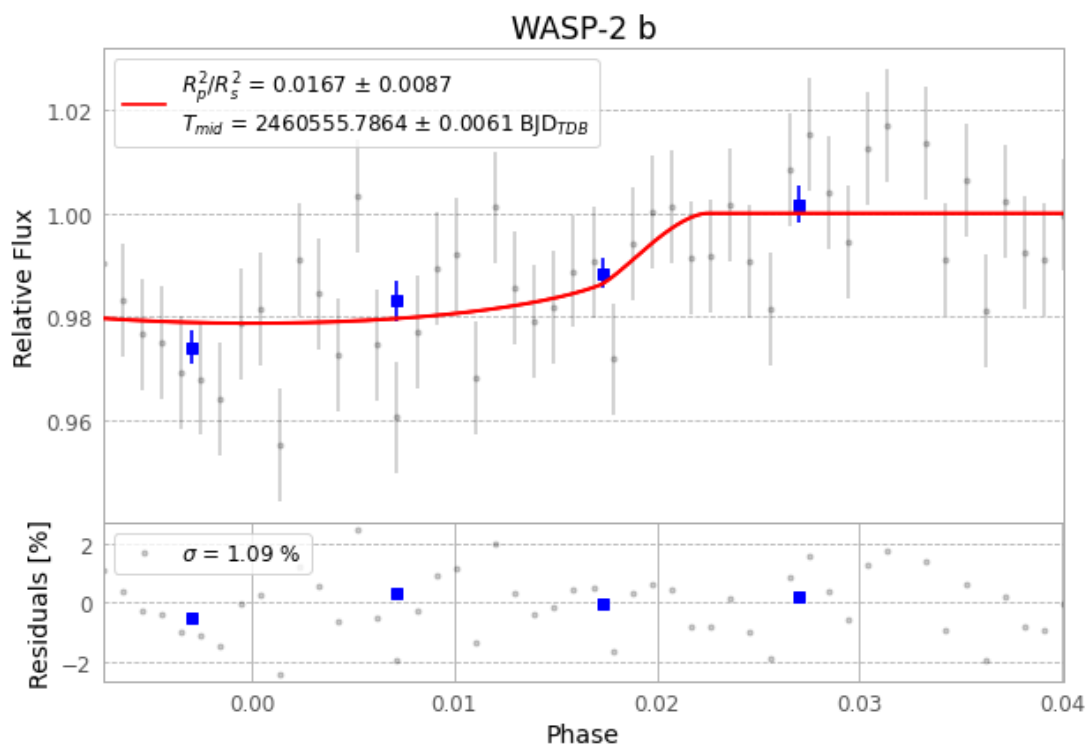


Figure 1 — FinalLightCurve_WASP-2 b_2024-09-01.png (SHA-256 33ee69a9cbc06fd6...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [308, 254], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 f7284ea92cd52cf8...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 825eb7e

Data provenance

50 FITS frames (33 MB), WASP-2240902062115.FITS → WASP-2240902084815.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-2-240902.log (dbd82e6db1a6...), FinalLightCurve_WASP-2 b_2024-09-01.png (33ee69a9cbc0...), FinalLightCurve_WASP-2 b_2024-09-01.csv (301b3ff87719...)

Compute resources

Wall time 100.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-17 06:52Z.